

## Phase 2 RAG Retrieval

## Pre-requisite for Classwork

## UserStories Ingested

The screenshot shows the RAG Pipeline interface. On the left, the 'DATA PIPELINE' section includes 'Convert to JSON', 'Embeddings & Store', and 'SEARCH ENGINES' (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion). The main area displays the 'Convert User Stories' button and a 'Reset' button. Below, the 'Conversion Results' section shows a success message: 'Success! User stories file converted successfully'. The 'OUTPUT FILE' is 'stories-1789868765539.json' and the 'STATUS' is 'Completed'. The 'CONVERSION LOG' shows 'Converted 190 user stories from "stories" into stories-1789868765539.json'. A sidebar on the right lists labels: Labels, Sprint, Epic, Acceptance Criteria, Created Date, and Updated Date.

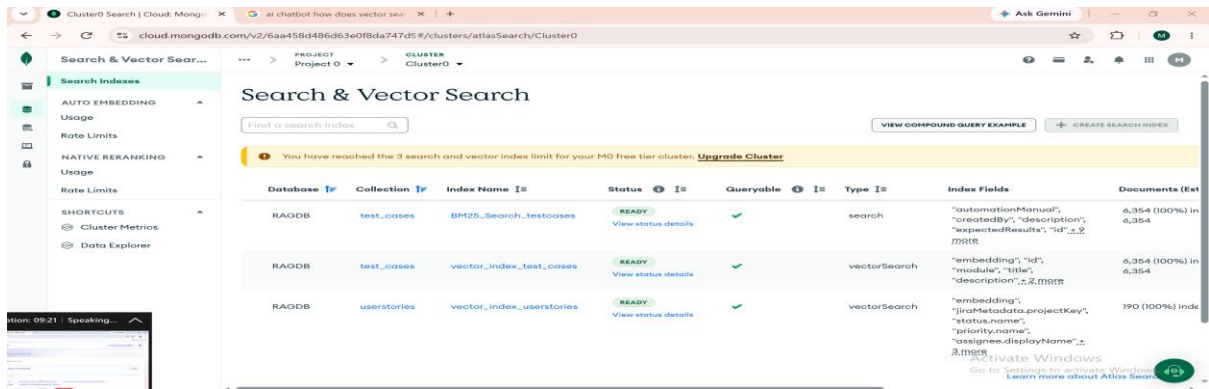
The screenshot shows the MongoDB Atlas interface. The 'Data Explorer' on the left shows the 'test\_cases' collection. The main area displays the 'Cluster0 > RAGDB > userstories' view. The 'Documents' tab shows a list of documents with fields like '\_id', 'key', 'summary', 'description', 'status', 'priority', 'acknowledged', 'reporter', 'created', 'updated', 'components', 'labels', 'flavorings', 'storypoints', and 'project'. The 'CONVERSION LOG' shows 'Converted 190 user stories from "stories" into stories-1789868765539.json'.

## TestCases Ingested

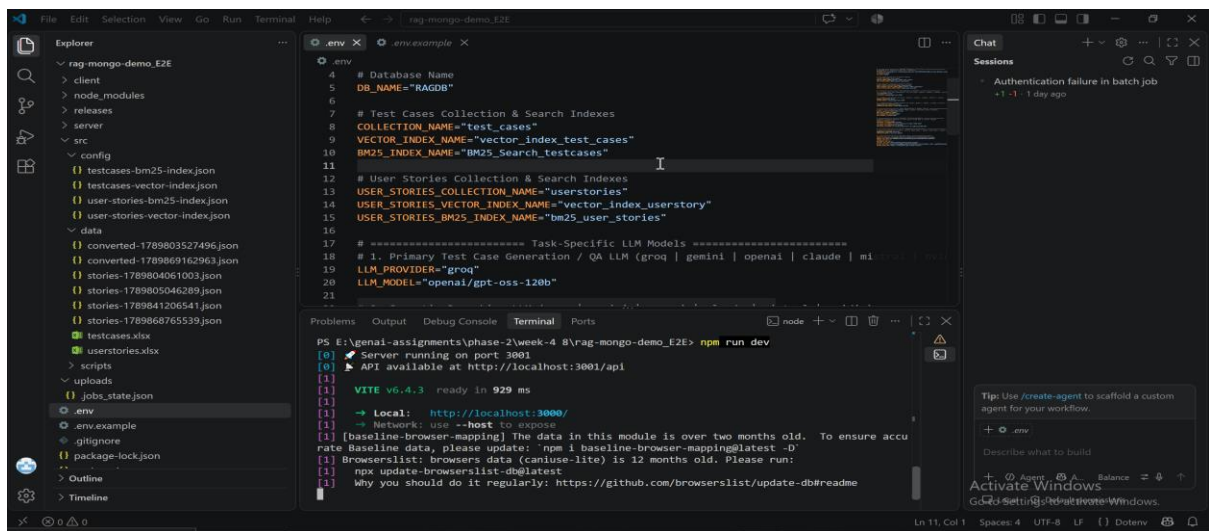
The screenshot shows the RAG Pipeline interface. On the left, the 'DATA PIPELINE' section includes 'Convert to JSON', 'Embeddings & Store', and 'SEARCH ENGINES' (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion). The main area displays the 'Convert Test Cases' button and a 'Reset' button. Below, the 'Conversion Results' section shows a success message: 'Success! Test cases file converted successfully'. The 'OUTPUT FILE' is 'converted-1789869162963.json' and the 'STATUS' is 'Completed'. The 'CONVERSION LOG' shows 'Converted 6354 test cases from "testcases" into converted-1789869162963.json'. A sidebar on the right lists labels: Last modified date, Risk, Version, and Type.

The screenshot shows the MongoDB Atlas interface. The 'Data Explorer' on the left shows the 'test\_cases' collection. The main area displays the 'Cluster0 > RAGDB > test\_cases' view. The 'Documents' tab shows a list of documents with fields like '\_id', 'module', 'id', 'preRequisites', 'title', 'description', 'steps', 'expectedResults', 'automationManual', 'priority', 'createdBy', 'createdDate', 'lastModifiedDate', and 'linkedStories'. The 'CONVERSION LOG' shows 'Converted 6354 test cases from "testcases" into converted-1789869162963.json'.

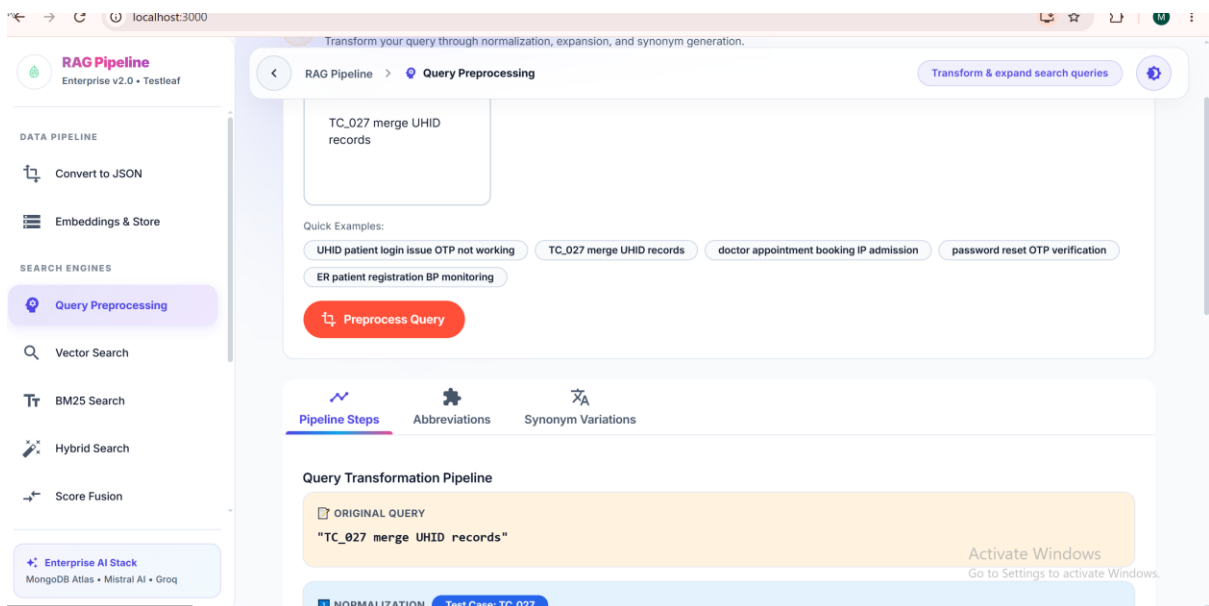
## Added BM25 in MongoDB using Atlas Search



## Updated .env file



## UI Query Preprocessing



## UI Query Preprocessing results

The screenshot displays the RAG Pipeline interface at localhost:3000. The left sidebar shows the 'Query Preprocessing' search engine selected. The main panel shows the preprocessing results for the query "test case 027 merge unique health id records".

**ABBREVIATION EXPANSION** (2 expansions)

Expanded:

- uhid → unique health id
- tc → test case

**SYNONYM EXPANSION** (5 variations)

Multiple query variations generated for comprehensive search coverage

- "TC\_027 test case 027 merge unique health id records"
- "TC\_027 examination case 027 merge unique health id records"
- "TC\_027 investigation case 027 merge unique health id records"

+ 2 more variations

**PROCESSING METADATA**

| Processing Time | Token Count | Abbreviations | Synonyms |
|-----------------|-------------|---------------|----------|
| 201ms           | 5           | 2             | 2        |

Activate Windows  
Go to Settings to activate Windows.

## Vector Search

The screenshot displays the RAG Pipeline interface at localhost:3000. The left sidebar shows the 'Vector Search' search engine selected. The main panel shows the vector search results for the query "Share Diagnostic Reports with Patients via WhatsApp".

**Share Diagnostic Reports with Patients via WhatsApp** (Clear)

Use descriptive terms related to test functionality

**TRY AN EXAMPLE QUERY**

- Share Diagnostic Reports with Patients via...
- Login test cases for the Billing module wi...
- Validate payment gateway timeout and retry...
- Password reset flow across web and mobile
- High-risk regression scenarios for checkou...

Advanced Metadata Filters ☐ Show Filters

Results Limit: 5 Results

**Search Query:** "Share Diagnostic Reports with Patients via WhatsApp" (5 results)

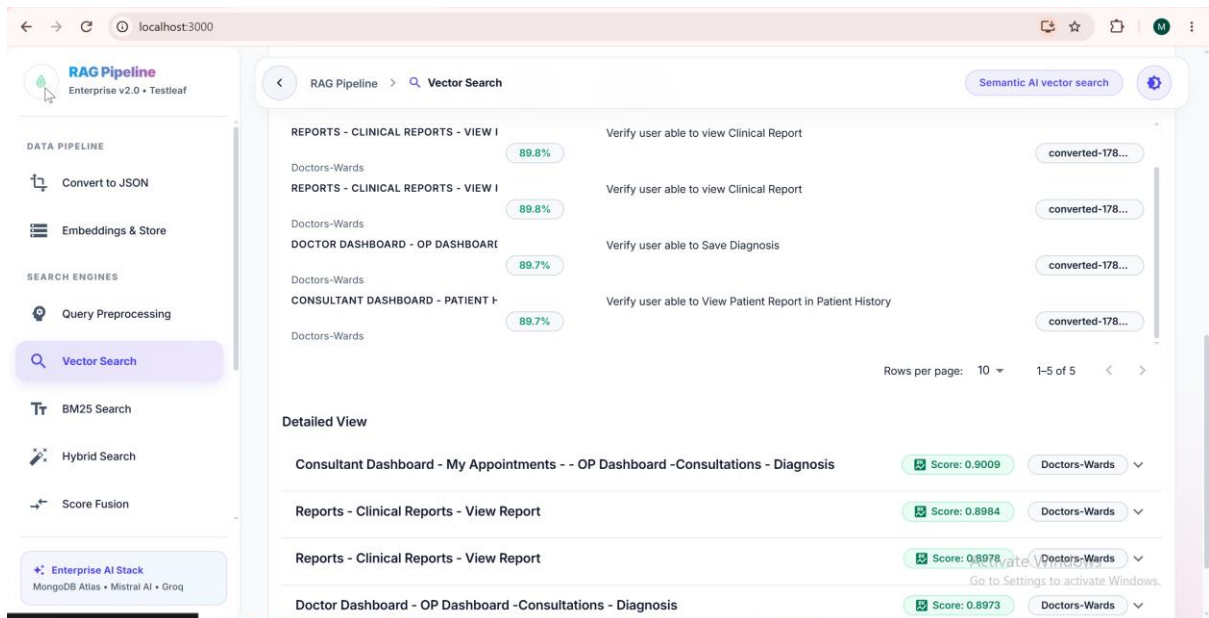
Performance: \$0.000001 cost 14 tokens

**Search Results**

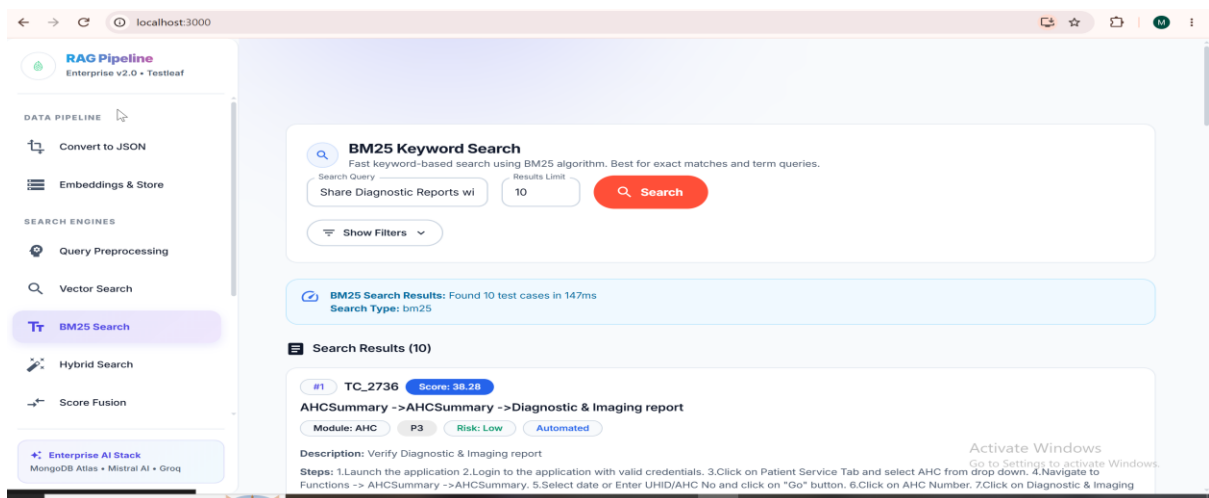
5 results ranked by semantic similarity

| Test Case | Similarity | Description | Source |
|-----------|------------|-------------|--------|
|-----------|------------|-------------|--------|

Activate Windows  
Go to Settings to activate Windows.



## BM25 Keyword Search



## Hybrid Search

